Tanishq Srivastava

Software Developer

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Summary

Computer Science undergraduate focused on backend systems. Built layered REST APIs in TypeScript, Node.js, and PostgreSQL with secure session authentication and transactional checkout that prevents overselling under concurrent requests. Shipped ML-backed services in Python and FastAPI with model serving, experiment tracking, and data-drift monitoring. Solved 500+ DSA problems across various competitive programming platforms; strong in OOP, DBMS, Operating Systems, and Computer Networks.

Education

ABES Engineering College — Ghaziabad, Delhi NCR
B.Tech. in Computer Science and Engineering — *Aug 2024 – 2028*

Technical Skills

Languages	TypeScript, JavaScript, C++, Python, SQL
Backend	Node.js, Express.js, FastAPI, REST API design
Databases	PostgreSQL, MySQL
Frontend	React.js, Tailwind CSS, HTML5, CSS3
Tools	Git, GitHub, Docker
ML	PyTorch, scikit-learn, XGBoost, SHAP, MLflow, Evidently AI
Core CS	Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks

Projects

Acheter – Transaction-Safe Commerce Application GitHub
React, TypeScript, Express.js, PostgreSQL, Argon2id, Zod

- Built a full-stack platform with a Route–Controller–Service architecture and 12 REST endpoints covering authentication, products, carts, checkout, orders, and health checks over a 6-table PostgreSQL schema.
- Engineered transactional checkout and concurrency-safe inventory using PostgreSQL transactions, conditional stock deduction, and rollback across orders, order items, and carts, preventing overselling during competing checkouts.
- Secured sessions with 32-byte random tokens stored as SHA-256 hashes, Argon2id password hashing, HttpOnly cookies, and Zod request validation.

Loan Default Prediction – Credit Risk Platform GitHub
Python, XGBoost, SHAP, FastAPI, PostgreSQL, MLflow, Streamlit, Docker

- Built an end-to-end credit-risk pipeline on 150,000 records, benchmarking 6 model configurations (balanced and tuned variants) across Logistic Regression, Random Forest, and XGBoost.
- Reached 0.8693 validation ROC-AUC with XGBoost; served predictions via a FastAPI service (health, predict, history, drift endpoints) backed by PostgreSQL, with a Streamlit UI.
- Containerized with Docker Compose; tracked experiments in MLflow, explained predictions with SHAP, and monitored feature drift with Evidently AI.

RadSync – Explainable Chest X-Ray Analysis GitHub
PyTorch, DenseNet121, Grad-CAM, FastAPI, React

- Trained a DenseNet121 multi-label classifier in PyTorch for five thoracic conditions, achieving 0.87 F1-score and 96% test accuracy.
- Added Grad-CAM to produce class-specific localization heatmaps, making model predictions interpretable.
- Built a FastAPI inference backend and React/TypeScript dashboard for X-ray upload, prediction visualization, historical case context, and downloadable reports.

Achievements

- **Hackathon Finalist:** Selected as a Finalist among 700+ competing teams in HackSynergy 2026.
- **Solved over 500 DSA problems** in C++ among various platforms such as Leetcode, Codeforces, GeeksForGeeks etc.